

MARITIME COMPLIANCE TOOLS

USER GUIDE

Canadian CII Calculator — Web Version



Carbon Intensity Indicator • Attained / Required CII • Transport Canada GHG Reporting Framework

Prepared for: Vessel Compliance & Reporting Teams

Document version: 1.0

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1 Introduction

1.1 Purpose of this guide

This guide explains how to use the Canadian CII (AER) Calculator, a browser-based tool for calculating a vessel's Carbon Intensity Indicator (CII) under the Annual Efficiency Ratio (AER) method, in line with Transport Canada's vessel GHG reporting framework. It walks through every input field, the fuel consumption table, the results, rating band and multi-year trend, using screenshots taken directly from the tool.

The guide is written for ship operators, technical superintendents, and compliance officers who need to enter voyage and fuel data, run a CII calculation, interpret the resulting rating, and produce a shareable report.

1.2 What the tool does

- Calculates total CO₂ emissions from fuel consumption entries, using carbon conversion factors aligned with the IMO's Fourth GHG Study.
- Calculates the Attained CII (AER) and compares it against the Required CII for the vessel's ship type, capacity and reporting year.
- Assigns a CII rating band (A–E) under the Canadian schedule and shows where the vessel sits on the rating scale.
- Plots an interactive multi-year CII trend chart against the rating boundaries, and tabulates Required CII and rating for each year.
- Imports/exports fuel data via Excel, saves/reloads full cases as XML, and produces a print-ready PDF/Word compliance report.

Before you start

- The tool runs entirely in your browser — no data is uploaded or sent anywhere.
- Have the vessel's Bunker Delivery Notes (BDNs), voyage records/logbook distance data, and IMO particulars ready.
- Use a current version of Chrome, Edge, or Firefox for the best experience.

2 Interface Overview

The calculator is organised into a fixed toolbar and three numbered sections: A — Vessel & Year Inputs, B — Fuel Consumption, and C — Results. Each section can be collapsed using the arrow in its header.

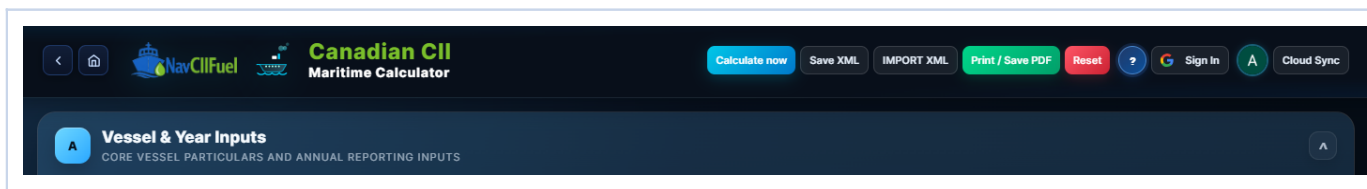


Figure 1 — Toolbar: Calculate now, Save XML, Import XML, Print / Save PDF, and Reset.

The toolbar buttons are available at all times and are described in the table below.

Button	Function
Calculate now	Runs the CII/AER calculation using the current Part A and Part B inputs and reveals Section C.
Save XML	Exports all current inputs (vessel data and fuel rows) as an .xml case file for later reuse.
Import XML	Loads a previously saved .xml case file and repopulates all fields automatically.
Print / Save PDF	Opens a formatted, print-ready compliance report and the browser print dialog (choose "Save as PDF" as the destination).
Reset	Clears all entered data and returns the calculator to a blank state.
? (Page Help)	Opens the built-in on-screen help guide describing every field in Sections A–C.

3 Quick Start Workflow

Follow these five steps for a typical single-vessel, single-year CII calculation:

Step	What to do
1	Open the calculator HTML file in your browser and confirm the page loads with Sections A and B visible.
2	In Section A, enter the vessel name, IMO number, ship type, reporting year, capacity, distance sailed and reporting period.
3	In Section B, add one row per fuel type consumed and enter the tonnes consumed for each, from BDNs and fuel records.
4	Click Calculate now and review the CO ₂ emitted, Attained CII, Required CII, rating band and multi-year trend in Section C.
5	Use Save XML to archive the case and Print / Save PDF to generate the formal report for distribution.

4 Section A — Vessel & Year Inputs

Section A captures the vessel identity and the parameters that define the scope of the calculation: reporting year, capacity, distance sailed and reporting period.

Figure 2 — Section A with sample vessel data entered.

Field	Description
Vessel Name	Free-text identifier for the vessel. Appears on the printed report; does not affect calculations.
IMO	The vessel's unique 7-digit IMO identification number. Appears on the report only — does not affect calculations.
Ship Type	Selects the vessel category (e.g. Bulk Carrier, Tanker, Container Ship, Gas Carrier). Determines the a and c coefficients used to compute the Required CII, and whether capacity is measured in DWT or GT.
Year	The reporting year. Each year carries a progressively stricter required CII reduction factor under the Canadian schedule.
Capacity	The vessel's deadweight tonnage (DWT) or gross tonnage (GT), depending on ship type. This is the denominator in the AER formula: $AER = CO_2(g) \div (Capacity \times Distance)$.
Distance (nm)	Total distance sailed during the reporting year, in nautical miles, from the vessel's official voyage records or logbook.
Reporting Period (From / To)	Start and end dates of the annual reporting period (typically 1 Jan – 31 Dec). Appears on the printed report and is used to auto-fill the Year field. Both dates must fall within the same calendar year.
Tip	- The Capacity unit (DWT/GT) is determined by the selected Ship Type — always re-check the capacity value after switching ship types. - Keep the Reporting Period dates within the same calendar year as the selected Year field.

5 Section B — Fuel Consumption

Section B records every fuel type consumed during the reporting year. CO₂ for each row is calculated as Consumption (t) × CF (tCO₂/tFuel).

FUEL	CF (tCO ₂ /tFUEL)	CONSUMPTION (T)	ACTIONS
Heavy fuel oil (HFO)	3.114	3800	Remove
Diesel/gas oil	3.206	420	Remove

+ Add Row

Figure 3 — Fuel Consumption table with two fuel rows entered (HFO and MGO).

Field / Control	Description
Fuel (column)	Fuel type selected from the drop-down (HFO, LFO, MDO, MGO, LNG, methanol, ethanol, LPG and more). Choose "Other" for a fuel not listed and enter its name manually.
CF (tCO₂/tFuel)	Carbon conversion factor for the selected fuel, aligned with the IMO's Fourth GHG Study. Auto-filled for standard fuels; enter manually for custom fuels.
Consumption (t)	Total mass of the fuel consumed in the reporting year, in metric tonnes, sourced from Bunker Delivery Notes and fuel records.
Add Row / Remove	Add Row appends a new blank fuel row; Remove (per row) deletes that row. There is no limit on the number of rows.
Excel Template	Downloads a blank .xlsx file with the correct column headers (Fuel, CF, Consumption) for offline data entry.
Upload Excel	Imports a completed .xlsx, .xls or .csv file and populates all fuel rows automatically, matching fuel names to the built-in list.
Clear All	Removes all fuel rows and replaces them with a single blank row — use to start the fuel section over.
Blend / Biofuel	For biofuel blends, quantity can be entered by mass (MT) or volume (m ³); density is used to convert volume to mass where applicable.
Note	This Canadian edition of the calculator does not include the optional correction-factor table (FCvoyage, TF, FCElectrical, etc.) found in the IMO edition — CO₂ and distance are used as entered.

6 Section C — Results

Section C appears after Calculate now is clicked. It requires no direct input — every value is derived automatically from Sections A and B, following the Canadian CII (AER) methodology under Transport Canada's vessel GHG reporting framework.

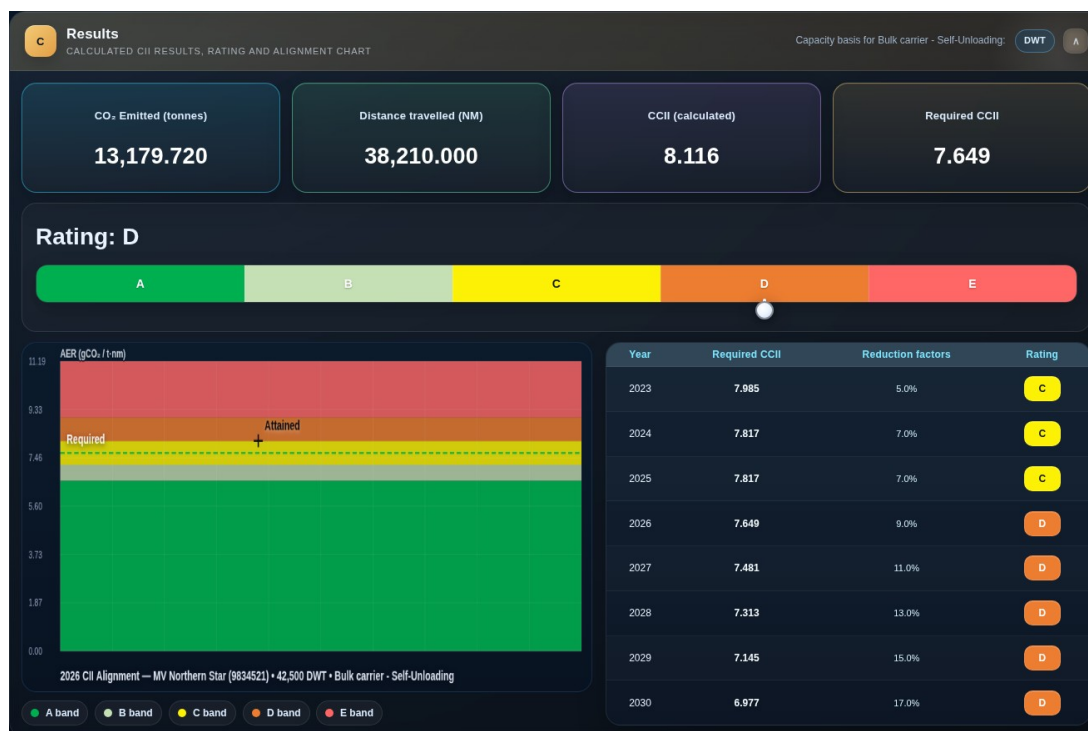


Figure 4 — Calculated KPIs, rating band and CII trend chart for the sample vessel.

Result	Meaning
CO₂ Emitted (t)	Total CO ₂ emissions for the reporting year: $\Sigma (\text{Fuel} \times \text{CF})$ across all fuel rows in Section B.
Distance (NM)	The distance sailed during the reporting year, as entered in Section A — the second denominator in the AER formula.
Attained CII (AER)	$\text{AER} = \text{CO}_2 (\text{g}) \div (\text{Capacity (t)} \times \text{Distance (nm)})$. This is the single official figure reported under Canada's CII regime, in gCO ₂ /(t·nm).
Required CII	The vessel's Required CII for the selected year, derived from the ship-type coefficients a and c and the year's mandatory reduction factor under the Canadian schedule.
Rating (A-E)	The vessel's Canadian CII rating band, from comparing Attained AER to the Required CII boundaries: A = Superior · B = Above average · C = Average (compliant) · D = Below average (corrective action plan required if held 3 consecutive years) · E = Inferior (immediate corrective action plan required).
CII Trend Chart	Plots Attained AER against the Required CII boundary lines for bands A-E across the full reporting horizon, with the selected year highlighted. Hover over the chart to read exact values.
Year Trend Table	Tabulates Required CII, Reduction factor and Rating for each year, based on current Section A/B inputs — useful for planning compliance ahead of stricter annual targets.

7 Exporting and Record-Keeping

- Save XML after completing each case, to preserve every input value for future reuse or audit.
- Import XML to reload a saved case rather than re-entering data.
- Use Print / Save PDF once the results have been checked, choosing "Save as PDF" in the browser print dialog.
- Keep an editable Word copy of the report alongside the PDF for internal review and version tracking.
- Name saved files consistently, e.g. VesselName_IMO_ReportingYear_CII.xml / .pdf, for easy retrieval.

8 Best-Practice Checklist

- Use the same reporting year and reporting period consistently in Section A.
- Confirm capacity is entered in the correct unit (DWT or GT) for the selected ship type.
- Enter fuel consumption in tonnes and double-check CF values before calculating, especially for custom or biofuel entries.
- After every calculation, review the rating band and Required CII before exporting the report.
- Store the XML input file together with the generated PDF/Word report for full traceability.

9 Troubleshooting

Symptom	Likely cause / fix
Section C does not appear	Calculate now has not been run yet, or a required field in Section A is empty — fill all fields and click Calculate now.
Capacity unit is DWT when GT was expected (or vice versa)	The Ship Type does not match the vessel — re-select the correct ship type; the capacity basis updates automatically.
Uploaded Excel file did not populate all rows	Fuel names in the file do not match the built-in list — unmatched fuels are set to "Other"; check and correct CF values manually.
Chart or trend table looks outdated	Run Calculate now again after changing Year or Ship Type, as these affect the Required CII line.
Print / Save PDF opens a blank or partial report	Allow the browser a few seconds to render before printing, and ensure pop-ups are not blocked for this file.
Section A/B changes don't seem to update the results	Section C automatically hides when Section A or B inputs change — click Calculate now again to refresh it.

A Appendix — Full Interface Reference

The overview below shows the complete calculator page with sample data entered and a calculation already run, for reference alongside the section-by-section descriptions in this guide.

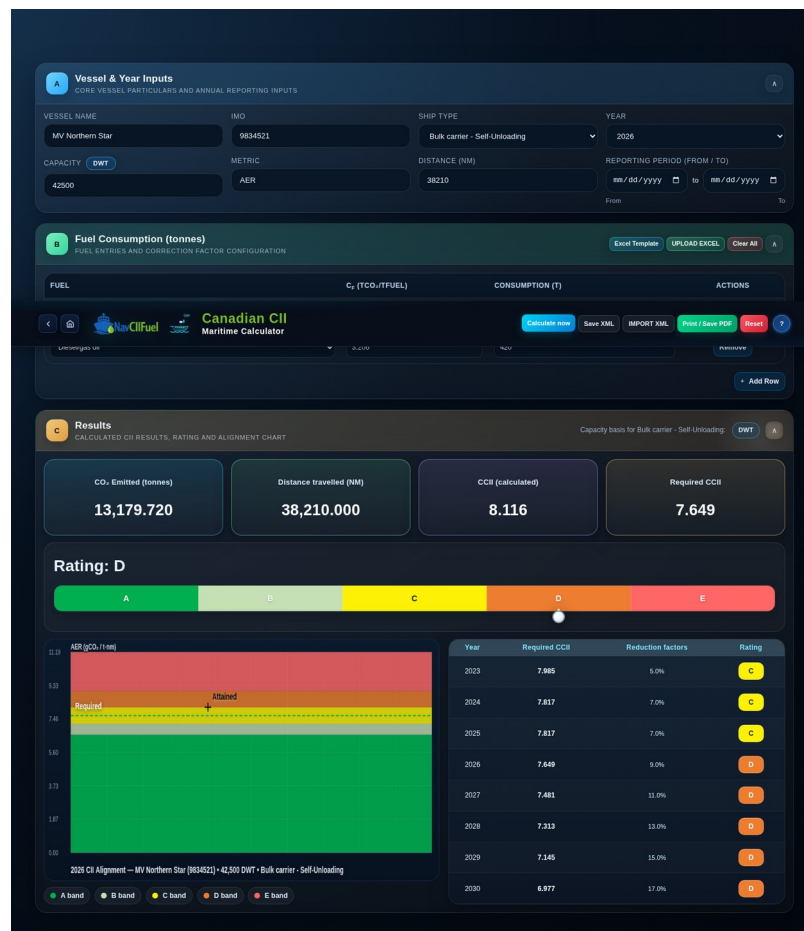


Figure 5 — Full calculator page after data entry and calculation.

End of user guide.